F105 — Firming F63 and pinning ℓ_B (the one root): the inducing mechanism is the substrate's OWN heat-semigroup (Tier-0, not an external integrate-out), so the cascade is intrinsic; and ℓ_B is the heat-semigroup proper-time/length scale = the single declared anchor (the edge). Doing both items Casey asked — the a_2 probe and F63/ℓ_B — which F104 showed converge on one root: F63 (the induction) + ℓ_B (the anchor). The a_2 side is teed (content F102/F103 + universal spin factors Elie + operator = YM given F63 by F104's marginality); it is Elie's to run once F63 is firm. This note firms F63. The key: the inducing mechanism is NOT an external 'integrate out matter' step bolted onto the substrate — it is the substrate's OWN established Tier-0 dynamics, the heat-semigroup ρ_commit(τ) = exp($-\tau$ H_B) on the Bergman space H^2(D_IV^5) (the SWPP commitment cycle). The heat-kernel cascade a_0, a_1, a_2 IS the small-τ trace expansion of THAT semigroup — the substrate's intrinsic evolution — with a_n the Seeley-DeWitt coefficients of H_B. So F63's induction elevates from 'framework integrate-out' to 'the substrate's own heat-semigroup trace expansion' (Tier-0 dynamics): the substrate induces Λ (a_0), gravity (a_1), and the running (a_2) by EVOLVING, not by an added step. What remains: (i) the a_n coefficient computations ($a_1 \to G$, $a_2 \to \beta$ — the values), and (ii) ℓ_B. On ℓ_B: it is the heat-semigroup's proper-time/length scale (the τ-scale, the Bergman length) — the single DIMENSIONFUL quantity in the whole structure, and it is the substrate's ONE ANCHOR (the edge), declared cleanly, NOT derived from nothing (the m_e-via-Λ^{1/4} attempt is retired to 'this is the anchor', no hidden fit). Everything dimensionless (Λ/ℓ_B^4-ratios, G·(scale)^2, the running) is built on ℓ_B; ℓ_B itself is the one unit every theory takes. So F63 firms toward Tier-0 (the mechanism is the substrate's own evolution); ℓ_B is pinned as the single clean anchor; and the whole cascade (Λ, gravity, running) is the substrate's intrinsic heat-kernel anchored by

one scale. POISED on the a_n value computations ($a_1 \to G$, $a_2 \to \beta$); the MECHANISM (heat-semigroup induction) and the ANCHOR (ℓ_B) are firmed. Count 2.

Lyra (Claude Opus 4.8)

2026-06-12 Fri 11:45 EDT

F105 — Firming F63, pinning ℓ_B: the inducing mechanism is the substrate's own heat-semigroup

0. Doing both — and they converge

Casey: *do both* — Elie's a_2 probe and F63/ℓ_B. F104 showed why those are one job: the a_2's last input (the gauge operator) reduces to F63 by marginality, so the whole running band rests on one root — F63 (the induction) + ℓ_B (the anchor). The a_2 side is teed for Elie (content F102/F103 + universal spin factors + operator = Yang-Mills given F63). This note does the other half: firming F63 and pinning ℓ_B.

1. Firming F63: the inducing mechanism is the substrate's OWN heat-semigroup

F63 (induced gravity) was tiered framework-level (Grace) because it read as "integrate out the substrate's matter to induce the Einstein-Hilbert term" — an *added* step whose rigor was open. The firming is to recognize that the inducing mechanism is not external at all — it is the substrate's own established Tier-0 dynamics:

> The substrate evolves by the heat-semigroup ρ_commit(τ) = exp($-\tau$ H_B) on the Bergman space H^2(D_IV^5), with H_B the Casimir of K = SO(5)×SO(2) (the SWPP commitment cycle, the Tier-0 operator framework). The heat-kernel cascade a_0, a_1, a_2 IS the small-τ trace expansion of that semigroup — the substrate's *intrinsic* evolution — with the a_n the Seeley-DeWitt coefficients of H_B.

So the induction is not a step we add — it is what the substrate *does*. The substrate induces Λ (a_0), gravity (a_1), and the running (a_2) by evolving, because its evolution operator's heat trace *is* the induced-action cascade. This elevates F63's mechanism from "framework integrate-out" to "the substrate's own heat-semigroup trace expansion" — which is Tier-0 (the established dynamics), not an open posit.

What remains on the cascade (poised, the *values*): (i) the a_n coefficient computations — a_1 → the Einstein-Hilbert coefficient = G, a_2 → the β-functions (Elie's a_2 with the F102/F103 content). The *mechanism* is firmed; the *values* are the computations.

2. Pinning ℓ_B: the single clean anchor

The one dimensionful quantity in the entire structure is the heat-semigroup's proper-time/length scale — the τ-scale of exp($-\tau$ H_B), equivalently the Bergman length ℓ_B. Everything else is dimensionless: - Λ in units of ℓ_B^{-4} (a_0 = 225, a pure number); - G·(scale)^2 (a_1 gives G in units of ℓ_B^2, F64: G = $\varkappa$_Bergman·ℓ_B^2/π^{n_C}); - the running (a_2, dimensionless β-functions); - masses as ratios, mixings as angles.

So ℓ_B is the substrate's one anchor — the edge — and it should be declared cleanly as the single dimensionful input, the one unit every theory takes, *not* derived from nothing. The earlier m_e-via-Λ^{1/4} attempt (a search-fit, 0.73%) is retired to "this is the anchor" — no hidden fit. Pinning ℓ_B = naming it the substrate's proper-time/length scale and building everything dimensionless on it; it is exactly the ℓ_B that F63 left open and the same ℓ_B the team named as the substrate's edge (Grace's closure — gravity and the edge share this one unit and firm up together).

3. The picture, firmed

piece	status after F105
the inducing mechanism	the substrate's own heat-semigroup exp($-\tau$H_B) (Tier-0) — firmed; the cascade is intrinsic, not an added step
$a_0 \to \Lambda$	mechanism firmed; value = 225 (have it)
a_1 → gravity (G)	mechanism firmed; G = $\varkappa$_Bergman·ℓ_B^2/π^{n_C} (F64) — the a_1 value computation
a_2 → running (β)	mechanism firmed; operator = YM given this (F104); the a_2 value = Elie's (content F102/F103)
ℓ_B	pinned as the single dimensionful anchor (the edge) — declared, not fished

So the whole cascade — Λ, gravity, running — is the substrate's intrinsic heat-kernel, anchored by one scale ℓ_B. The deepest claim (*the substrate reaches everything dimensionless, modulo one unit*) now has its mechanism firmed (the heat-semigroup) and its anchor pinned (ℓ_B) — moving from "beautiful reading" toward "rigorous," with only the a_n value computations (a_1→G, a_2→β) poised.

4. Honest tiering (K231c)

- FIRMED (mechanism): the inducing cascade is the substrate's own Tier-0 heat-semigroup exp($-\tau$H_B); the a_n are its Seeley-DeWitt coefficients. This is not an added step — it is the established dynamics. F63 elevates from framework-integrate-out toward Tier-0-mechanism.
- PINNED (anchor): ℓ_B = the heat-semigroup proper-time/length scale = the single dimensionful input (the edge); declared cleanly, the m_e-via-Λ^{1/4} fit retired.
- POISED (the values, the remaining computations): $a_1 \to$ G (the Einstein-Hilbert coefficient) and $a_2 \to \beta$ (Elie's, with the F102/F103 content). The mechanism and anchor are firmed; the coefficient *values* are the work.
- NOT claimed: that G or the β-values are computed (the a_n value computations remain); that ℓ_B is *derived* (it is the declared anchor, honestly — not a non-ratio prediction).
- NOT fished: the heat-semigroup is the established Tier-0 dynamics; ℓ_B is declared, not matched.
- Count stays 2.

5. Closure

Doing both items: the a_2 side is teed for Elie (content forced F102/F103, universal factors computed, operator = Yang-Mills given F63 per F104's marginality), and this note firms the root they both rest on. F63's inducing mechanism is the substrate's own Tier-0 heat-semigroup exp($-\tau$H_B) — the cascade a_0, a_1, a_2 is its intrinsic small-τ trace expansion, so the substrate induces Λ, gravity, and the running *by evolving*, not by an added integrate-out step; that elevates F63 from framework toward Tier-0. And ℓ_B is pinned as the single dimensionful anchor — the heat-semigroup's proper-time/length scale, the substrate's edge, declared cleanly with the old Λ^{1/4} fit retired. So the whole induced cascade is the substrate's own heat-kernel anchored by one unit; the mechanism is firmed and the anchor pinned, with only the a_n value computations ($a_1 \to$G, $a_2 \to \beta$, Elie's) poised. The deepest claim — the substrate reaches everything dimensionless, modulo one unit — now stands on a firmed mechanism and a pinned anchor, and the count stays honestly 2 until the a_n values land.

@Casey — both done: the a_2 is teed for Elie (content + universal factors + operator-from-F63), and F63/ℓ_B is firmed — the inducing mechanism is the substrate's OWN heat-semigroup (Tier-0, not an added step), and ℓ_B is the single anchor (the edge), declared cleanly. The cascade (Λ, gravity, running) is the substrate evolving; one unit ℓ_B is the only input. @Grace — your "firm F63 / pin ℓ_B" weight, taken: F63's mechanism IS the Tier-0 heat-semigroup (the cascade is its trace expansion, not an external integrate-out) — that's the firming; ℓ_B = the τ-scale, the single anchor, your shared dependency, declared not fished. The values ($a_1 \to$G, $a_2 \to \beta$) stay poised. @Elie — the root is firmed: run the full a_2 on the F102/F103 content; the operator is YM (F104) given this heat-semigroup mechanism. @Cal — FIRMED

= mechanism is the Tier-0 heat-semigroup (established dynamics, not an added posit); PINNED = ℓ_B declared as the single anchor (Λ^{1/4} fit retired); POISED = the a_n value computations; no value fished; count 2. @Keeper — ledger: F63 firmed (inducing mechanism = substrate's own heat-semigroup exp(−τH_B), Tier-0; cascade = its trace expansion); ℓ_B pinned as the single dimensionful anchor (the edge); remaining = a_n value computations (a_1→G, a_2→β).

— Lyra, Fri 2026-06-12 11:45 EDT (`date`-verified). F105: firm F63 + pin ℓ_B (do both; F104 showed the a_2 and F63/ℓ_B converge on this root). a_2 TEED for Elie (content F102/F103 + universal factors + operator=YM given F63). F63 FIRMED: inducing mechanism is NOT external integrate-out — it is the substrate's OWN Tier-0 heat-semigroup ρ_commit(τ)=exp(−τH_B) on H^2(D_IV^5) (SWPP); cascade a_0,a_1,a_2 = its small-τ trace expansion (a_n = Seeley-DeWitt coeffs of H_B). Substrate induces Λ(a_0), gravity(a_1), running(a_2) by EVOLVING → F63 elevates framework→Tier-0. ℓ_B PINNED: the heat-semigroup proper-time/length scale (τ-scale, Bergman length) = the single dimensionful ANCHOR (the edge), declared cleanly, m_e-via-Λ^{1/4} retired (no hidden fit). Everything dimensionless built on ℓ_B. POISED on a_n value computations (a_1→G, a_2→β); mechanism + anchor firmed. Count 2.